⛈️ The Strike & Staying Safe: Reading

Chemistry · Unit 1C · Level 3 (closer to the class work) · goes with L1.10-L1.14 (Energy Flow, The Strike, Conductors & Insulators, No Swimming, Unit Test)

Name: _____

The Strike and Staying Safe

When charge separates in a storm cloud, energy is stored, a bit like a stretched rubber band. This is called potential energy. The more charge that separates, the more energy is stored. When the charge finally moves, the stored energy changes into heat (thermal energy), light and sound.

A strike happens in steps. First, a faint, zig-zag channel of negative charge called a stepped leader moves down from the cloud. As it gets close, positive streamers reach up from tall objects like trees, towers and people. When a leader and a streamer meet, a huge current flows. That bright flash is the return stroke. It heats the air so fast that the air expands with a boom: thunder.

Some materials let charge move through them easily. These are conductors. Metals are good conductors because some of their electrons are free to move from atom to atom. Other materials, like rubber, plastic, glass and dry wood, hold their electrons tightly. These are insulators.

Pure water does not conduct well, but salt water does. When salt dissolves, it breaks into ions: charged particles that can move. In the No Swimming lab, the light bulb got brighter as more salt was added, because more ions carried more current. Pools, lakes and the ocean all have dissolved ions, so lightning can travel through them.

That is why some places are safer than others. A lightning rod is a metal conductor that gives the charge a safe path into the ground. Buildings with wiring and plumbing, and hard-top cars, are safe places. Open fields, water and tall, lone trees are dangerous. Remember: when thunder roars, go indoors.

Check your reading

1. What energy changes happen during a lightning strike?

2. Why is salt water a conductor but pure water is not?

3. Explain how a lightning rod protects a house.